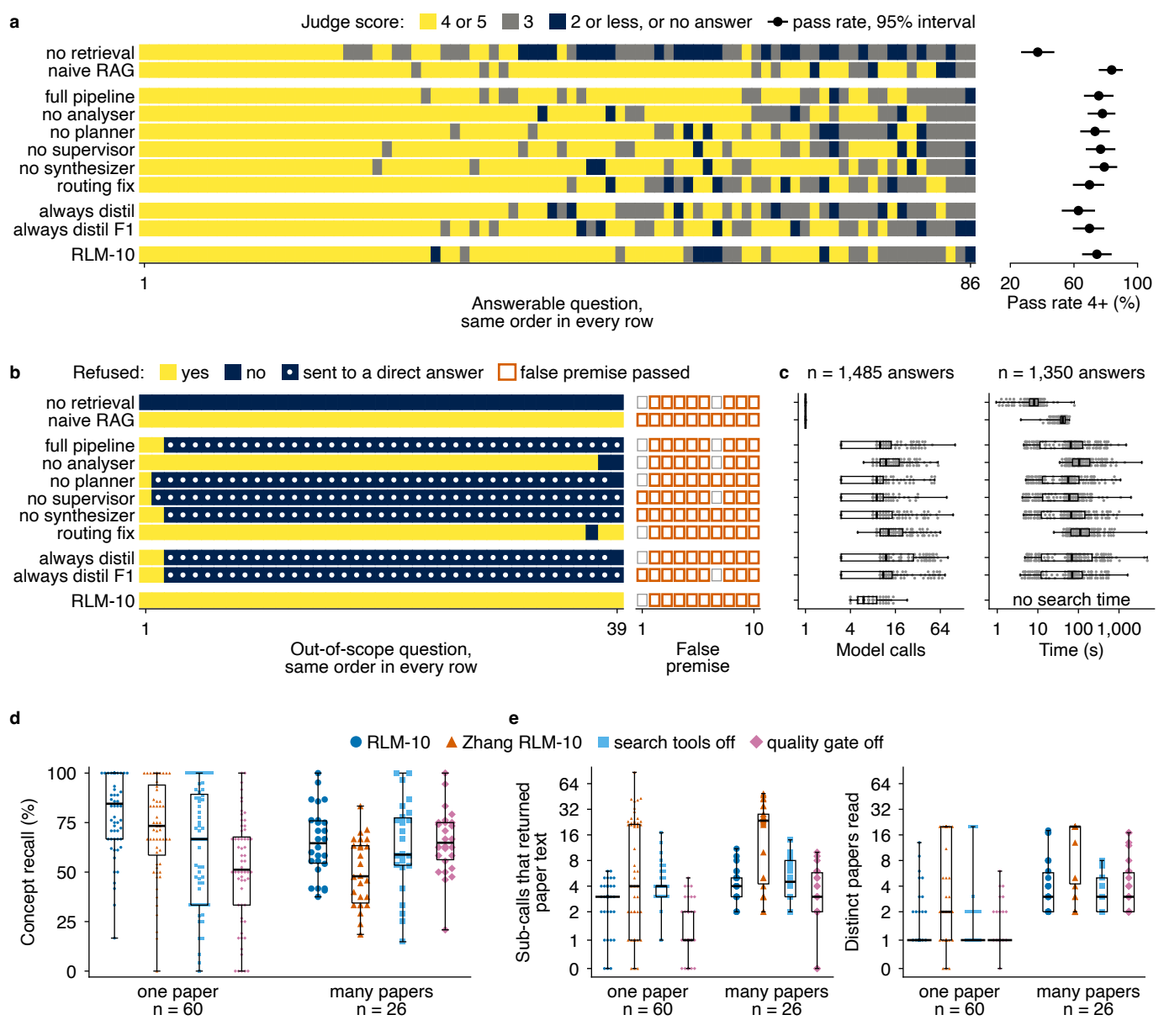

a
Judge score: 4 or 5 3 2 or less, or no answer pass rate, 95% interval
no retrieval
naive RAG
full pipeline
no analyser
no planner
no supervisor
no synthesizer
routing fix
always distil
always distil F1
RLM-10
1
86
Answerable question, same order in every row
20
60
100
Pass rate 4+ (%)
b
Refused: yes no sent to a direct answer false premise passed
1
39
Out-of-scope question, same order in every row
1
10
False premise
c
n = 1,485 answers
1
4
16
64
Model calls
n = 1,350 answers
no search time
1
10
100
1,000
Time (s)
d
RLM-10 Zhang RLM-10 search tools off quality gate off
Concept recall (%)
100
75
50
25
0
one paper
n = 60
many papers
n = 26
e
Sub-calls that returned paper text
64
32
16
8
4
2
1
0
one paper
n = 60
many papers
n = 26
Distinct papers read
64
32
16
8
4
2
1
0
one paper
n = 60
many papers
n = 26